Jie He

Advanced – R/SAS; Proficient – Python/SQL/Shell Scripting

RESEARCH INTERESTS

Design of clinical trials and development of analytic methods, to create computational frameworks and tools that enable statisticians and medical professionals to work effectively with longitudinal, multicenter or large-scale high-dimensional datasets in drug development and medical research.

EDUCATION

2024 Fall **Boston University**, Boston, MA
Ph.D. Student in Biostatistics:
Research assistant externship on SDTM and CSR automation at Vertex Pharmaceuticals.
Analysis of mobile health data (>1 billion observations) from older adults [1], advised by Prof. Chunyu Liu.
[R, SAS, and shell scripting; longitudinal analysis; sampling-based models]

2015-2020 **University of North Carolina at Chapel Hill**, Chapel Hill, NC
M.S., Biostatistics, *Not Providing Point-scaled Grades*
B.S., Biostatistics with a second major in Computer Science and a Math minor, *Distinction*
Thesis: *Weighted inference of gene expression variability in single cell RNAseq data for gene set tests: a replication study.* Advised by Prof. Di Wu. [R; high-dimensional scRNA-seq data; permutation-based models]

PROFESSIONAL EXPERIENCE

2021-2024 **Boston Children's Hospital**, Boston, MA
Biostatistician – Biostatistics and Research Department, supervised by Dr. Edie Weller

- Simulation study: Using simulated structures based on real-world electronic health records, applied and evaluated machine learning classification models on binary endpoints to assess performance across varying data types and correlation strengths.
- Prediction: Employed random survival forests on clustered data to identify key features predicting in-hospital mortality following ECPR, and to assess the prediction accuracy of current ELSO registry data.
- Survival analysis: Used cox models with frailty on clustered right-censored data for association analysis [2, 3, 4]. Also validated the theoretical framework of optimizing partial likelihood for time-dependent covariates [6].
- Data management tool: Developed an R package with custom functions and a Shiny app to streamline the data cleaning and descriptive statistics presentation of extensive, yet unorganized, patient data containing clinical notes entered in the REDCap database. Automated quality control of the data entries [7].

ACADEMIC PROJECTS

Fall 2020 **University of North Carolina at Chapel Hill**, Chapel Hill, NC
Graduate Research Assistant – Biostatistics, supervised by Dr. Donglin Zeng

- Merged national data of social determinants with timelines of COVID-19 policies (e.g., shutdowns, public mask mandates) and temporal-spatial U.S. mobility data from Google Maps, to support a doctoral researcher in incorporating these variables as covariates in time-series trend prediction models.

2018-2019 **University of North Carolina at Chapel Hill**, Chapel Hill, NC
Undergraduate Honors Thesis – Biostatistics, advised by Dr. Di Wu

- Developed and implemented R functions to model mean–variance relationships in zero-inflated single-cell RNA-seq data (32,738 genes × 2,692 cells) and conducted hypothesis testing for differential gene expression using gene-level variability as the test statistic.

2017-2018 **University of North Carolina at Chapel Hill**, Chapel Hill, NC
Undergraduate Research Assistant – Biomedical Research Imaging Center (BRIC), Drs. Dinggang Shen & Dong Nie

- Applied statistical and machine learning methods in Python (Keras) to develop convolutional neural network models for classifying brain glioma grades from MRI data (n=128), including model evaluation and reproducibility checks, to explore imaging-based biomarkers relevant to clinical research.

2017 **University of North Carolina at Chapel Hill**, Chapel Hill, NC

Working with Data in a Public Health Research Setting, Katherine J. Roggenkamp

- Reproduced results from the METS randomized controlled trial on metformin's weight loss and metabolic control effects using SAS macros to clean and analyze public datasets, perform statistical comparisons, and generate publication-ready outputs.

AWARDS

2019 **Undergraduate Honors Thesis with Distinction, UNC**

2018 **Phi Beta Kappa, the Phi Beta Kappa Society**

2018 **Carolina Research Scholar Award, UNC**

2017 **GHC Student Scholarship, AnitaB.org** (Grace Hopper Celebration of Women in Computing)

2013 **Second Prize at National Olympiad in Informatics (NOI), China** (Language Used: C++)

PUBLICATIONS & PRESENTATIONS

Accepted or In Review

2025 [1] He, J., et al. Associations Between Smartwatch-Derived Measures and Cognitive Function: Findings from the Electronic Framingham Heart Study. In Review 2025.

Full Publications

2025 [2] He J. Associations Between Smartwatch-Derived Measures and Cognitive Function: Findings from the Electronic Framingham Heart Study. *Boston University Health Science Poster Session*, Boston, MA, 2025. [Poster]

[3] Soulsby, W.D., Olveda, R., He, J., Berbert, L., Weller, E., Barbour, K.E., Greenlund, K.J., Schanberg, L.E., von Scheven, E., Hersh, A. and Son, M.B.F., 2025. Racial Disparities and Achievement of the Low Lupus Disease Activity State: A CARRA Registry Study. *Arthritis Care & Research*, *77*(1), pp.38-49.

2024 [4] Andresen, F., He, J., Weller, E., Goldberg, B., Reilly, C.R., Nakano, T.A., Bertuch, A.A., Geddis, A.E., Joos, M., Coyne, K. and Brundige, K., 2024. Outcomes of Hematopoietic Stem Cell Transplantation for High-Risk Marrow Features without Malignancy in Shwachman-Diamond Syndrome. *Blood*, *144*, p.441.

[5] Myers, K., He, J., Goldberg, B., Reilly, C.R., Wan, Y., Zhu, X., Cesaro, S., Cipolli, M., Bertuch, A.A., O'Farrell, C. and Watanabe, K., 2024. Incidence of Myeloid Malignancy in Shwachman-Diamond Syndrome: An International Cohort Study. *Blood*, *144*(Supplement 1), pp.2703-2703.

[6] Hadjinicolaou, A., Briscoe Abath, C., Singh, A., Donatelli, S., Salussolia, C.L., Cohen, A.L., He, J., Gupta, N., Merchant, S., Zhang, B. and Olson, H., 2024. Timing the clinical onset of epileptic spasms in infantile epileptic spasms syndrome: A tertiary health center's experience. *Epilepsia*.

2023 [7] Mamolea, C., He, J., Weller, E., Hoskote, A., Ryan, K., Cooper, D., Bhaskar, P., Thiagarajan, R. and Vitali, S., 2023. 207: Factors and Outcomes Associated with Circuit Change in the Neonatal and Pediatric ECLS populations: An Analysis of the ELSO Registry. ASAIO Journal, 69(Supplement 3), p.54.

2022 [8] Maddux, A.B., Berbert, L., Young, C.C., Feldstein, L.R., Zambrano, L.D., Kucukak, S., Newhams, M.M., Miller, K., FitzGerald, M.M., He, J. and Halasa, N.B., 2022. Health impairments in children and adolescents after hospitalization for acute COVID-19 or MIS-C. *Pediatrics, 150*(3), p.e2022057798.